

-- Cox Regression - Sarcopenia Onset -----

i Definition : fnih
i Covariates : fixed
i Dairy : continuous
i Route : cc
i Interaction: none

-- Building survival dataset (CC) --

Events: 22 sarcopenia cases out of 839 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 839 rows | 21 events

-- Fitting Cox models (CC) --

Unadjusted formula: survival::Surv(time, event) ~ dairy_exposure
Adjusted formula : survival::Surv(time, event) ~ dairy_exposure + BMI +
education_level + and smoking_status + mvpa_min_day_f1 + diabetes_status

-- Events Per Variable (EPV) [cc]

i Events : 21
i Coefs : 8
i EPV : 2.6

-- C-index (Harrell's concordance) [cc]

i C-index : 0.846 (95% CI: 0.79-0.902)
i SE : 0.0288
??? C-index >= 0.70 - good discrimination.

-- Global model tests [cc]

i Likelihood ratio test : $\chi^2(8) = 21.17$, $p = 0.00671$ <- preferred
i Wald test : $\chi^2(8) = 14.34$, $p = 0.0733$
i Score (log-rank) test: $\chi^2(8) = 20.17$, $p = 0.0097$
i Total person-years: 58891.2

-- 7. Log-log survival plot | pass: parallel lines across log(time)

Log-log plot saved to 03_outputs/Cox/cc_fnih_fixed_continuous_27072026_1811

-- 8. Cox-predicted vs KM - calibration by dairy quartile

Cox vs KM calibration plot saved to

03_outputs/Cox/cc_fnih_fixed_continuous_27072026_1811

-- Kaplan-Meier plot by dairy category [cc]

i Continuous dairy split into quartiles for KM visualisation.

Ignoring unknown labels:

o colour : "Dairy quartile"

i Log-rank p-value: 0.226

KM plot saved to 03_outputs/Cox/cc_fnih_fixed_continuous_27072026_1811

-- Incidence rates by dairy quartile [cc]

quartile n_pts n_events person_years rate_per_1000py ci_low ci_high 1 Q1 (low)

210 5 14759.5 0.34 0.11 0.79 2 Q2 210 7 14630.0 0.48 0.19 0.99 3 Q3 209 1

14627.1 0.07 0.00 0.38 4 Q4 (high) 210 8 14874.6 0.54 0.23 1.06

Incidence table saved to 03_outputs/Cox/cc_fnih_fixed_continuous_27072026_1811

Model results saved to

03_outputs/Cox/cc_fnih_fixed_continuous_27072026_1811/cc_model_results.csv